

Akshay Kale

Production ML | LLM Applications | MLOps | Cloud Infrastructure

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EDUCATION

PH.D.

ARTIFICIAL INTELLIGENCE

UNIVERSITY OF NEBRASKA OMAHA

2019 -- 2024

M.S.

COMPUTER SCIENCE

UNIVERSITY OF NEBRASKA OMAHA

2016 -- 2019

PROGRAMMING

Python • SQL
Java • JavaScript
R • C • PHP

MACHINE LEARNING

Scikit-Learn • PyTorch
TensorFlow • Statsmodels
LLMs • RAG
LangChain • Hugging Face
Agents • MCP
Generative AI • Explainable AI (XAI)
Knowledge Graphs • NLP

CLOUD & DATA

Azure • Snowflake
Databricks • Modal

MLOPS

Docker • Dagster
MLflow • Git
REST APIs • Linux
Jupyter

DATA ENGINEERING

Pandas • Polars
NumPy • Feature Engineering
Data Pipelines

SELECTED HONORS

Best Visualization, Datapalooza 2019
Research Scholarship, University of
Nebraska
Advantage Scholarship, University of
Nebraska

SUMMARY

- Machine Learning Engineer with 2+ years building production AI systems and a Ph.D. in Artificial Intelligence.
- Experienced across the full ML lifecycle, from data engineering to deployment and monitoring.
- Built scalable predictive analytics and decision-support solutions for enterprise operations.
- Skilled in MLOps, LLMs, RAG, explainable AI, and cloud-based ML pipelines.

WORK EXPERIENCE

Machine Learning Engineer — KIEWIT CORPORATION

April 2024 -- Present

- Designed, deployed, and monitored **production machine learning models** that identified over **65% of safety incidents** across **400+** large-scale construction projects using historical and real-time operational data.
- Built end-to-end **machine learning pipelines** in **Python** using **Pandas, Polars, Scikit-Learn, Statsmodels, Snowflake, and SQL**, covering feature engineering, model training, deployment, monitoring, and performance evaluation.
- Developed scalable **MLOps** workflows using **Docker, Dagster**, and cloud infrastructure to automate data ingestion, model execution, reporting, and production deployments.
- Designed and deployed **LLM-powered applications** and **AI agents** to automate schedule analysis, enterprise knowledge retrieval, and operational decision-support workflows.
- Built interpretable **AI solutions** and executive dashboards using **Explainable AI (XAI)** techniques, enabling stakeholders to understand model predictions and operational risk.
- Led large-scale analytical studies on **schedule performance, labor productivity**, and **indirect cost forecasting**, delivering executive reports that informed enterprise planning and operational strategy.

RESEARCH EXPERIENCE

Research Assistant — UNIVERSITY OF NEBRASKA OMAHA

January 2017 -- April 2024

- Conducted research in **Explainable Artificial Intelligence (XAI)**, developing methodologies to improve the **transparency, interpretability**, and **trustworthiness** of machine learning systems.
- Proposed and evaluated novel approaches for **model interpretability**, integrating **SHAP, Partial Dependence Plots (PDP), Wasserstein Distance**, and **Formal Concept Analysis (FCA)** to compare and explain machine learning models.
- Designed scalable **machine learning pipelines** and **data engineering workflows**, integrating heterogeneous datasets exceeding **21M+ records** to support predictive maintenance research.
- Authored **4 peer-reviewed publications** spanning **XAI, human-in-the-loop AI, interactive visualization**, and **predictive analytics**, presenting findings at academic and technical conferences.